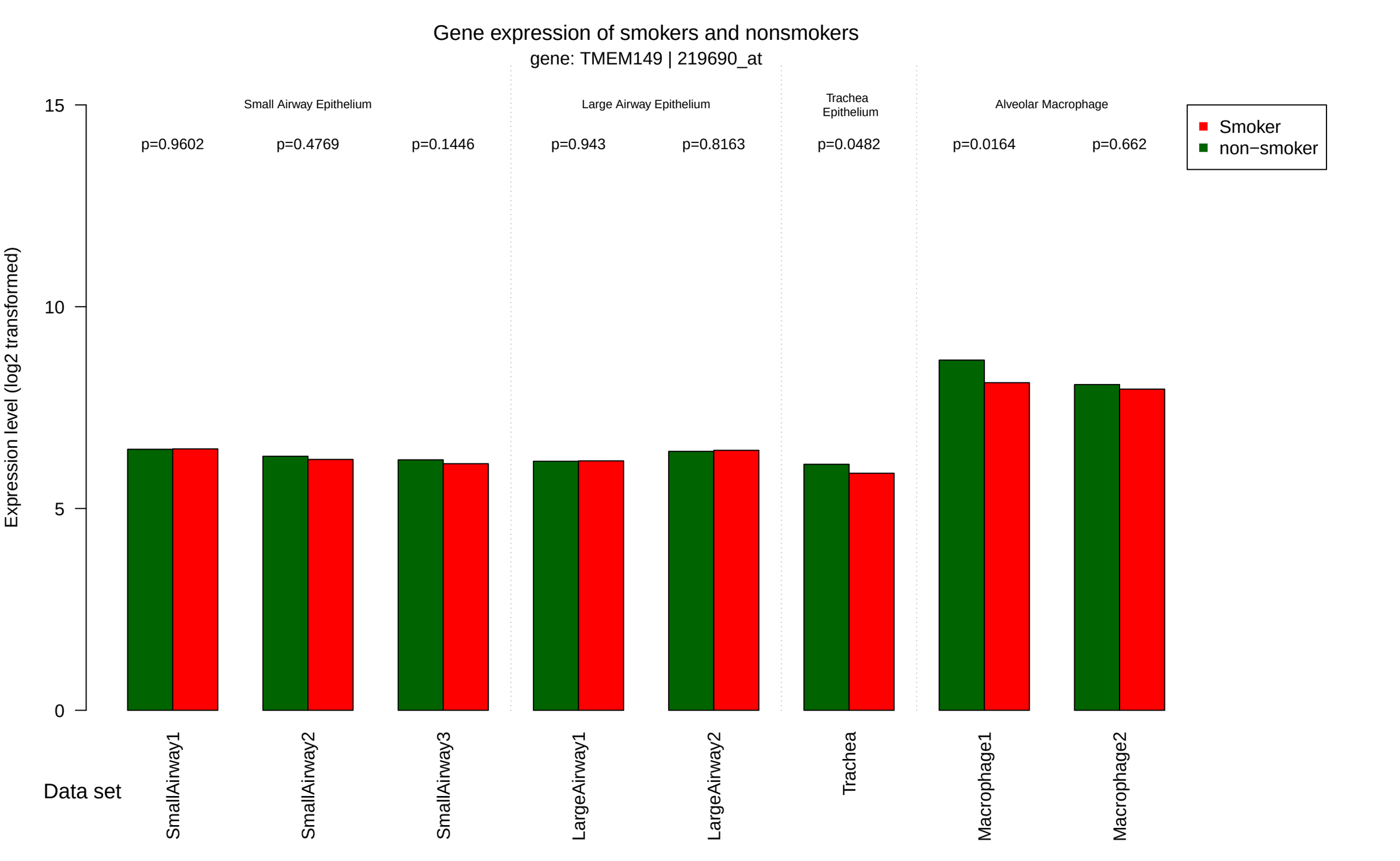
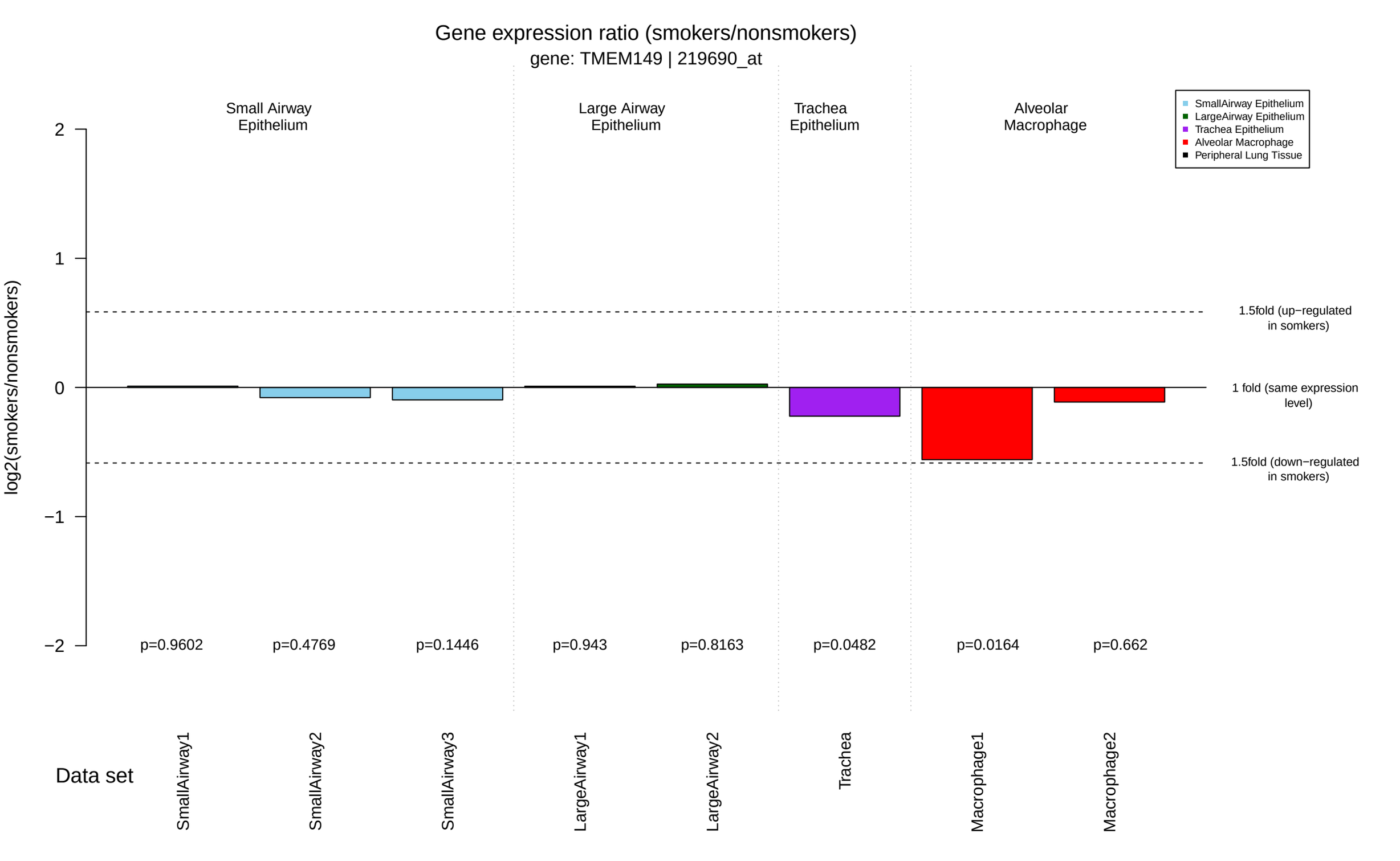
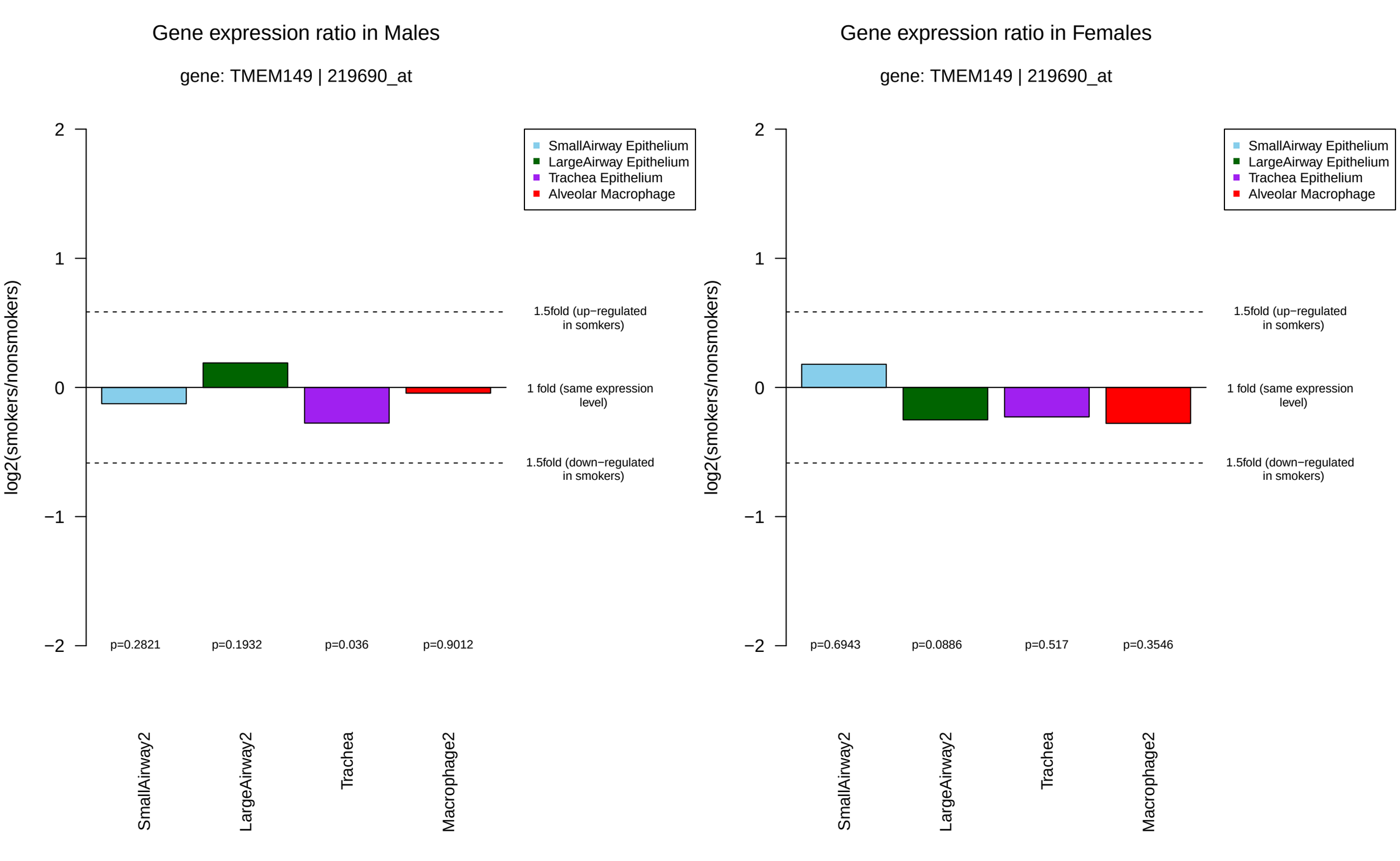
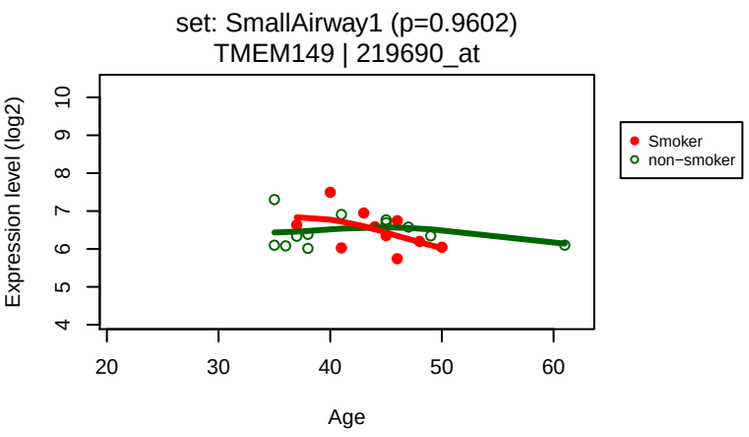
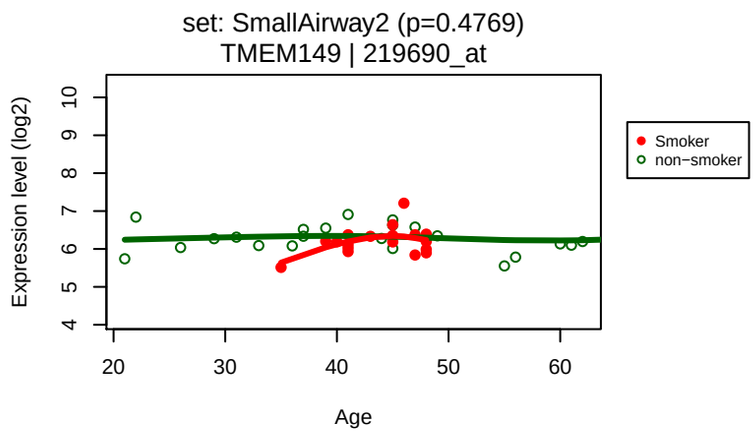
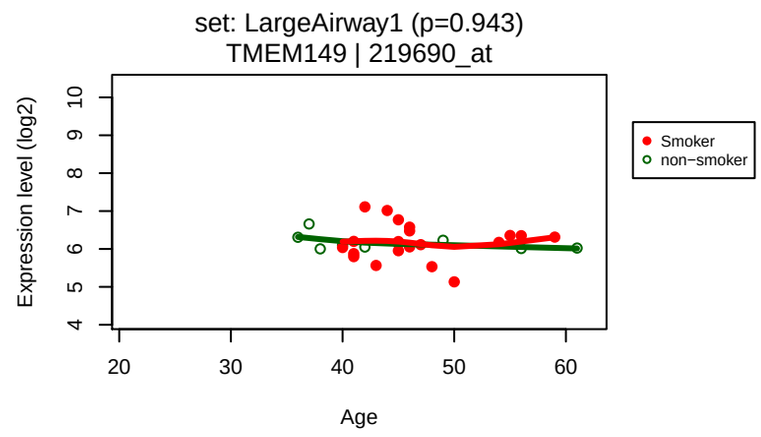

Gene expression of smokers and nonsmokers
gene: TMEM149 | 219690_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9602
p=0.4769
p=0.1446
p=0.943
p=0.8163
p=0.0482
p=0.0164
p=0.662
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

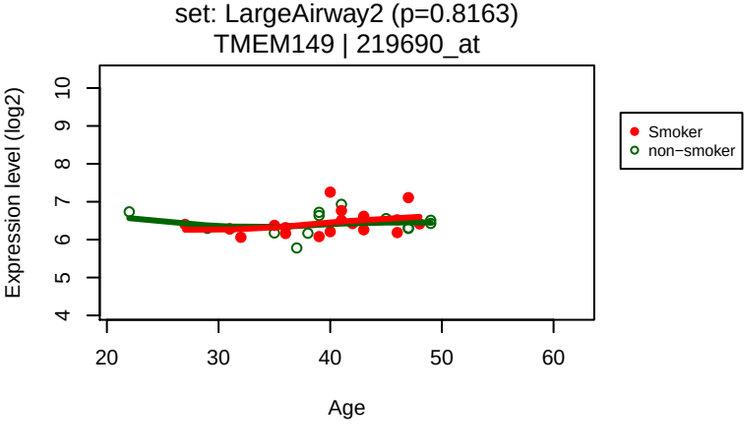

Gene expression ratio (smokers/nonsmokers)
gene: TMEM149 | 219690_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9602
p=0.4769
p=0.1446
p=0.943
p=0.8163
p=0.0482
p=0.0164
p=0.662
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

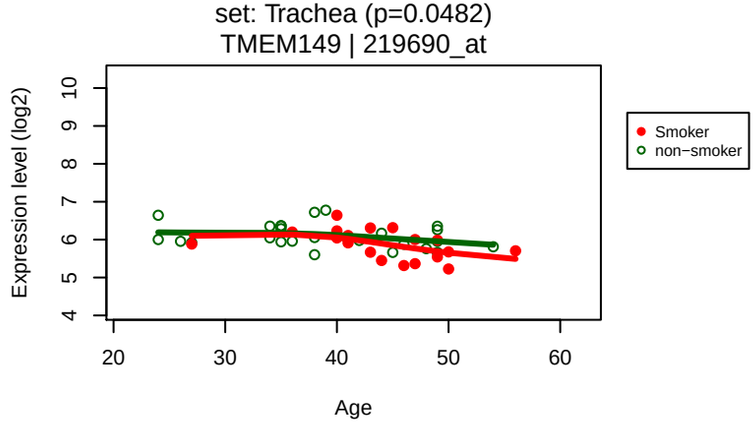

Gene expression ratio in Males
gene: TMEM149 | 219690_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2821
p=0.1932
p=0.036
p=0.9012
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM149 | 219690_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6943
p=0.0886
p=0.517
p=0.3546
SmallAirway2
LargeAirway2
Trachea
Macrophage2

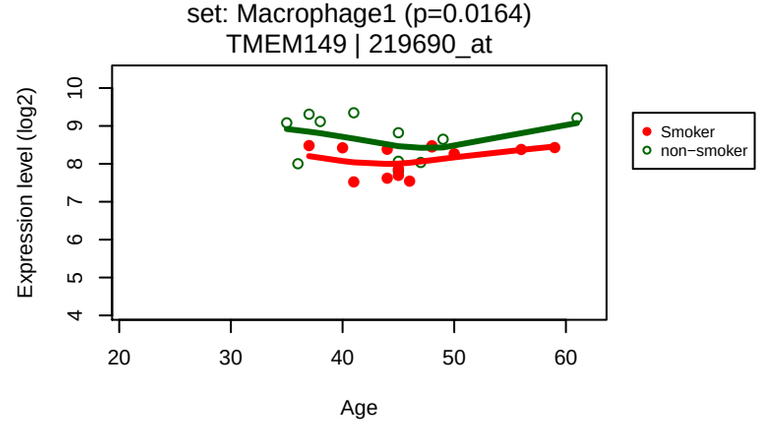

set: SmallAirway1 (p=0.9602)
TMEM149 | 219690_at
Expression level (log2)
Age
Smoker
non−smoker

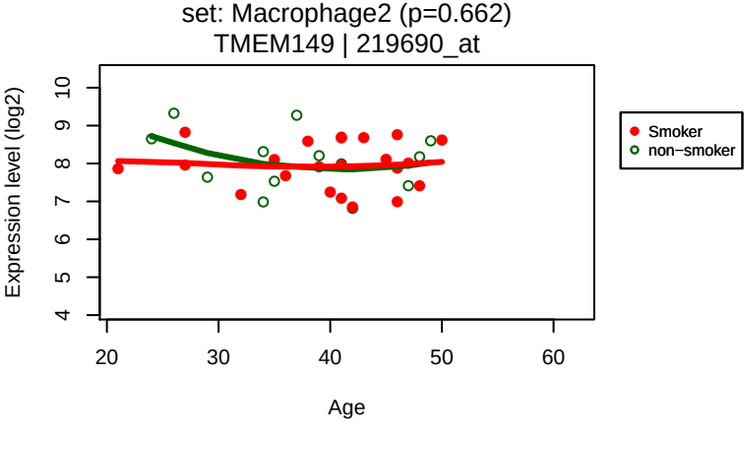

set: SmallAirway2 (p=0.4769)
TMEM149 | 219690_at
Expression level (log2)
Age
Smoker
non−smoker

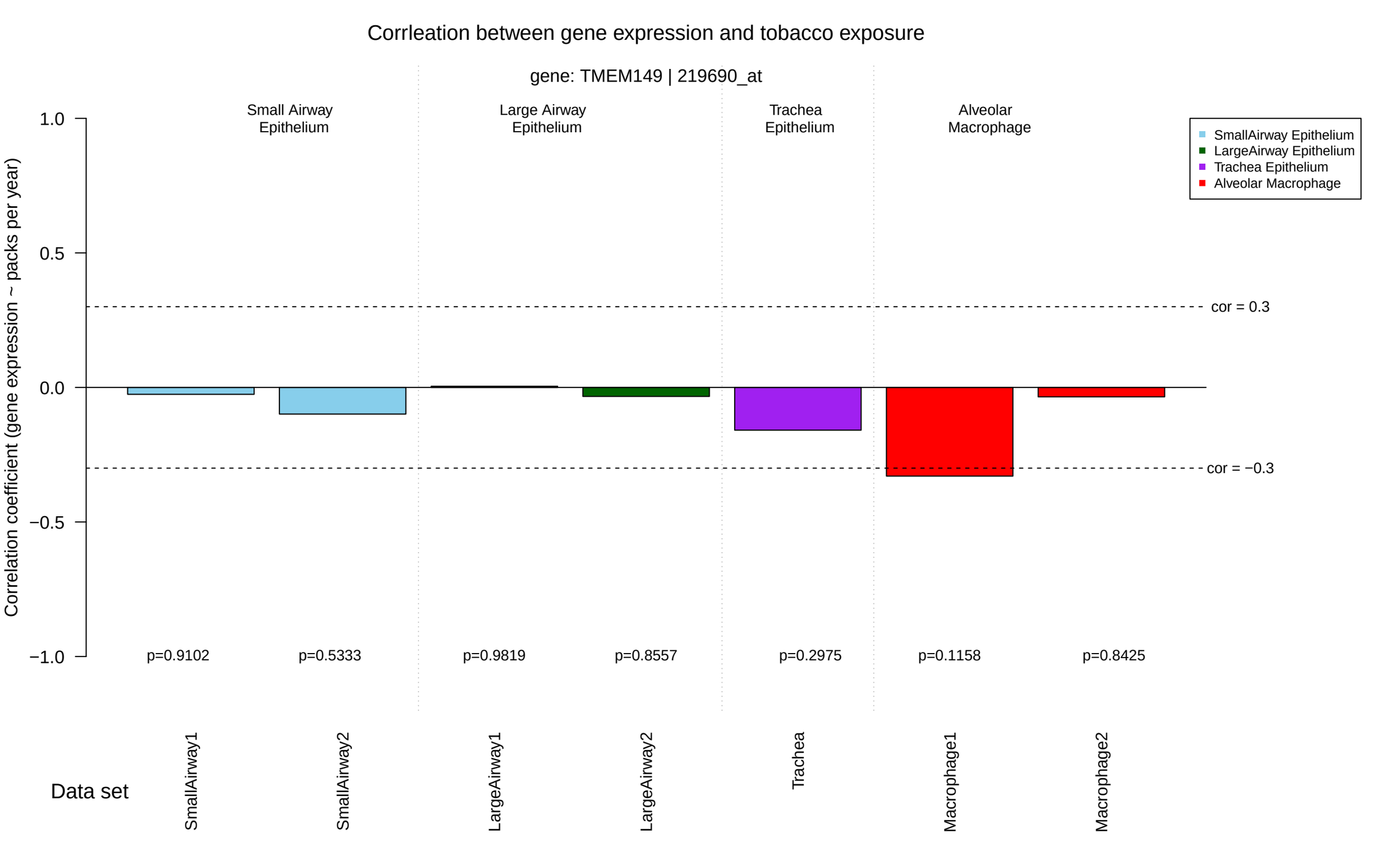

set: LargeAirway1 (p=0.943)
TMEM149 | 219690_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8163)
TMEM149 | 219690_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0482)
TMEM149 | 219690_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0164)
TMEM149 | 219690_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.662)
TMEM149 | 219690_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM149 | 219690_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9102
p=0.5333
p=0.9819
p=0.8557
p=0.2975
p=0.1158
p=0.8425
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2